

Synthetic Data is an Elegant GIFT for Continual Vision-Language Models

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https://github.com/Luo-Jiaming/GIFT_CL

Abstract

Pre-trained Vision-Language Models (VLMs) require Continual Learning (CL) to efficiently update their knowledge and adapt to various downstream tasks without retraining from scratch. However, for VLMs, in addition to the loss of knowledge previously learned from downstream tasks, pre-training knowledge is also corrupted during continual fine-tuning. This issue is exacerbated by the unavailability of original pre-training data, leaving VLM's generalization ability degrading. In this paper, we propose GIFT, a novel continual fine-tuning approach that utilizes synthetic data to overcome catastrophic forgetting in VLMs. Taking advantage of recent advances in text-to-image synthesis, we employ a pre-trained diffusion model to recreate both pre-training and learned downstream task data. In this way, the VLM can revisit previous knowledge through distillation on matching diffusion-generated images and corresponding text prompts. Leveraging the broad distribution and high alignment between synthetic image-text pairs in VLM's feature space, we propose a contrastive distillation loss along with an image-text alignment constraint. To further combat in-distribution overfitting and enhance distillation performance with limited amount of generated data, we incorporate adaptive weight consolidation, utilizing Fisher information from these synthetic image-text pairs and achieving a better stability-plasticity balance. Extensive experiments demonstrate that our method consistently outperforms previous state-of-the-art approaches across various settings.

1. Introduction

Vision-Language Models (VLMs), such as CLIP [43] and ALIGN [27], have offered unprecedented advancements in zero-shot generalization and fine-tuning performance on

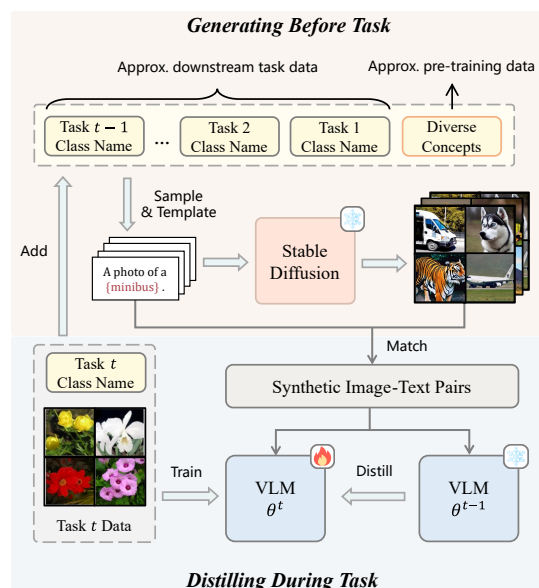


Figure 1. We use synthetic data generated by Stable Diffusion [46] to support continual fine-tuning of VLMs. By creating prompts from learned downstream class names and diverse visual concepts (i.e., additional class names), the generated data effectively approximates both downstream and VLM's pre-training data. During training, knowledge distillation [24] enables the VLM to re-experience previous knowledge via its past responses on matching synthetic images and corresponding text prompts.

various downstream tasks. Despite these advancements, difficulties remain in updating knowledge of pre-trained VLMs and applying them to multiple downstream tasks simultaneously. Vanilla approaches involve retraining VLMs from scratch to update knowledge or storing a fine-tuned VLM for each specific task, both of which incur substantial additional costs. While Continual Learning (CL) [6, 31, 34, 47, 48] can reduce this cost by enabling trained models to only learn new task data incrementally, thereby presenting itself as an efficient alternative to address these difficulties.

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Continual learning faces the fatal problem known as *catastrophic forgetting* [36], where a model loses previously acquired knowledge upon learning new tasks. For continually fine-tuned VLMs, in addition to forgetting learned downstream tasks, forgetting pre-training knowledge significantly impairs their zero-shot generalization ability. A straightforward approach to mitigate forgetting is to replay a small portion of stored historical samples [3, 6, 45], but this is impracticable for VLMs due to the inaccessibility of pre-training data. In parallel, recent advances in text-to-image generative models [44, 46] now enable synthesis of high-quality images from text descriptions, positioning synthetic data as a promising solution for training data scarcity [21, 52]. Studies [17, 29, 37] show that images generated by diffusion models [25, 46] can effectively approximate historical task data and mitigate forgetting in continual learning of randomly initialized convolutional networks. This promotes us to ask: *Can synthetic data from latest diffusion models help preserve pre-trained VLMs’ knowledge during continual learning, and if so, how?*

To answer this question, we first consider the following two sub-questions: (1) *How can diffusion model generate to approximate both the pre-training and downstream task data of VLMs?* (2) *How can the generated data be used to mitigate forgetting?* For question (1), we prompt a pre-trained Stable Diffusion [46] with class names from downstream tasks in continual learning to generate corresponding synthetic data. Additionally, findings in [67] suggest that VLM’s pre-training data can be well approximated by semantically rich external datasets like ImageNet [9]. This insight inspires us to generate pre-training synthetics in the same way as we generate for downstream tasks, but using diverse visual concepts as prompts. These concepts can be sampled from a corpus like the WordNet Synsets [38], or directly from ImageNet class names. As shown in Fig. 1, we overcome the challenge of inaccessible pre-training data and recreate historical samples without causing privacy and storage issues. Excessive overhead on image generation is generally undesirable. We achieve domain customization by generating images based on class names, enabling fewer generated images to encompass diverse distributions. However, the limited generated data volume still carries risks of in-distribution overfitting and reduced effectiveness in forgetting mitigation, necessitating a careful strategy for leveraging the limited synthetic data effectively. Therefore, as to question (2), we propose a novel continual learning framework for VLMs that leverages Generated data to Improve continual Fine-Tuning, named **GIFT**. GIFT comprises two core components: a knowledge distillation [24] process to preserve knowledge via synthetic image-text pairs, and an adaptive weight consolidation method for regularization.

Through knowledge distillation, the VLM is encouraged to mimic its original responses for synthetic samples and

thus preserve its well-learned feature representations. As a compressed, parameterized image library of diversity, Stable Diffusion shares part of its pre-training data space with the VLM. Its generated images and corresponding text prompts are widely distributed and highly aligned within the VLM’s feature space [23]. Based on this, we design a **contrastive distillation loss** similar to VLM’s pre-training objective of image-text matching, as well as an **image-text alignment constraint** to correct errors accumulated by the teacher model during the distillation process. Furthermore, we find that constraining parameter updates to a close distance from the pre-trained weights improves distillation effectiveness with limited synthetic data. This observation aligns with prior findings [51, 58] that combining fine-tuning with proper l_2 constraints can alleviate in-domain overfitting and maintain out-of-domain robustness, i.e., improving robustness to forgetting. Building on this, we propose **adaptive weight consolidation** for regularization. Unlike existing weight consolidation methods [2, 31, 66], our approach dynamically adjusts constraint levels on different parameters based on Fisher information [42] derived from synthetic data during training, thus preserving model stability without compromising plasticity. Our contributions are summarized as follows:

- We employ stable diffusion to recreate pre-training and historical samples for VLMs without causing privacy and storage issues, which helps the VLMs dynamically recall generic and learned downstream knowledge.
- We propose a novel continual fine-tuning framework GIFT that utilizes a small amount of synthetic data to assist VLMs in achieving a better balance between maintaining generalization and learning downstream tasks.
- Extensive experiments on 11 datasets across various domains, as well as traditional CL settings, demonstrate that our method consistently outperforms previous state-of-the-art approaches in various scenarios.

2. Related Works

Continual Learning (CL). CL aims to learn from sequential data, incorporating new knowledge without forgetting. Most CL approaches can be classified into three categories: architecture-based, regularization-based, and replay-based methods. Architecture-based methods [1, 14, 54, 61, 64] allocate new parameters for new tasks. The isolated parameters significantly reduce forgetting but also block knowledge transfer. Regularization-based methods try to mitigate forgetting by restricting parameter updates based on parameter importance [2, 31, 66], but limit model plasticity due to static regularization. Because these methods fail to adapt estimated parameter importance to the training dynamics, causing outdated and ineffective estimations as the task sequence progresses. We address this issue by leveraging Fisher information of real-time updated model pa-

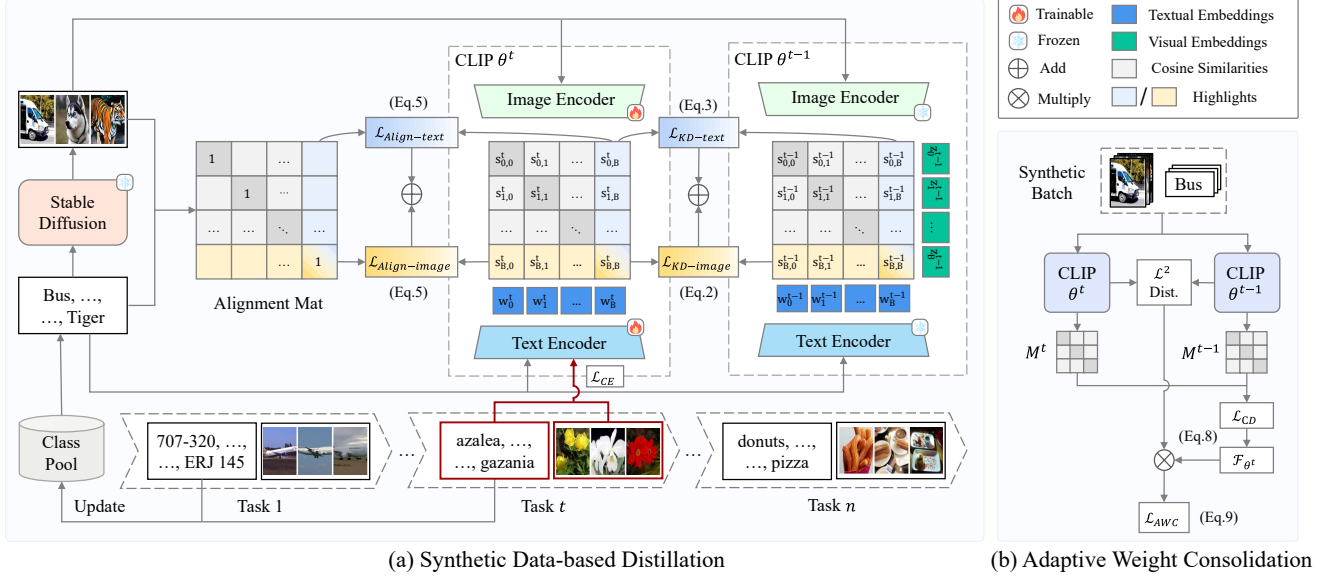


Figure 2. Framework overview of GIFT. (a) **Synthetic Data-based Distillation** aligns the output of the current CLIP model θ^t with the previous model θ^{t-1} on matching synthetic image-text pairs when learning a new task. Image-text alignment loss is applied to correct errors in the teacher model through hard target, i.e., the alignment matrix. (b) **Adaptive Weight Consolidation** employs a parameter importance weighted l_2 penalty to limit parameter changes causing forgetting and overfitting. By leveraging the Fisher information $\mathcal{F}_{\theta^t}$ from synthetic image-text pairs during training, parameter importance is adjusted in real-time to achieve a better stability-plasticity balance.

rameters on synthetic image-text pairs. Meanwhile, knowledge distillation [24] is typically used as a regularization term [11, 13, 34] to align the current output space with previous ones. Although the effect of knowledge distillation on current task dataset is often insufficient, distillation on stored historical samples achieves better performance when combined with experience rehearsal [6, 45]. Despite strong performance, replay-based methods [3, 4, 6, 45] that store partial historical samples are impractical in scenarios with storage limitations or privacy concerns.

Continual Learning with Synthetic Data. Some replay-based methods tackle the issue of inaccessible historical data by leveraging synthetic samples. Early works reproduce historical samples using model inversion [50, 63] or generative adversarial network (GAN) [20, 49, 59], but fall short due to high computational costs and low generation quality. Recent works have achieved success in overcoming forgetting with diffusion-generated data in continual learning of convolutional models [17, 29, 37] and continual object detection [30]. Taking advantage of the strong alignment between feature representations of diffusion-generated images and their text prompts, we introduce diffusion-generated data to the continual learning of VLMs.

Continual Learning for VLMs. Pre-training on large-scale image-text pairs endows Vision-Language Models (VLMs) [27, 34, 62] with strong generalization and even zero-shot capabilities. Extensive research has focused on fine-tuning VLMs to boost downstream task performance while preserving their inherent generalization abilities. Sev-

eral approaches optimize additional structures to avoid altering backbone parameters directly, such as prompt learning [28, 69, 70] and adapter tuning [16]. These fine-tuning strategies have been extended to CL settings by introducing task-specific structures [55–57, 65, 68]. However, the limited number of learnable parameters in these methods determines their suboptimal performance on complex tasks. Robust fine-tuning of model backbone parameters [19, 26, 51, 58] is also explored to achieve a stability-plasticity balance. To address continual learning scenarios with domain shift between tasks, ZSCL [67] enhances zero-shot capability protection through distillation on a large external reference dataset. We substitute the external reference dataset with a pre-trained generative model, which significantly reduces storage overhead while maintaining the diversity of the distillation data sources.

3. Method

3.1. Preliminaries

Continual Learning. Given a set of n tasks $\{\mathcal{T}^i\}_{i=1}^n$, continual training is conducted sequentially on each task $\mathcal{T}^i = (\mathcal{D}^i, \mathcal{C}^i)$. Here, $\mathcal{D}^i$ represents the task dataset $\{\mathbf{x}_j^i, \mathbf{y}_j^i\}_{j=1}^{N_i}$ with N_i instances, where $\mathbf{x}_j^i$ is an input image belonging to class $\mathbf{y}_j^i \in \mathcal{Y}^i$. And $\mathcal{Y}^i$ is the label space of the i^{th} task $\mathcal{T}^i$. Class names $\mathcal{C}^i = \{c_j^i\}_{j=1}^{|\mathcal{Y}^i|}$ maps the image labels to specific object names. The goal of continual learning is to maintain high performance across all tasks. We focus on two continual learning settings [53]. In Task-Incremental

Learning (TIL), the image $\mathbf{x}$ to be predicted is provided with its task identity t , so the model generates predictions within C^t . In Class-Incremental Learning (CIL), the task identity t is not given at inference and the model must distinguish between all the previously encountered classes $C = \bigcup_{i=1}^t C^i$.

Vision-Language Model. This paper focuses on Contrastive Language-Image Pre-training (CLIP) [43] as the VLM. During pre-training, CLIP jointly learns an image encoder $f_i(\cdot)$ and a text encoder $f_t(\cdot)$. Given an input image $\mathbf{x}$, the output distribution of CLIP is formulated as:

$$p(y_i | \mathbf{x}) = \frac{\exp(\cos(\mathbf{z}, \mathbf{w}_i) / \tau)}{\sum_{j=1}^{|Y|} \exp(\cos(\mathbf{z}, \mathbf{w}_j) / \tau)}, \quad (1)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity, τ is the temperature parameter learned during pre-training, $\mathbf{z} = f_i(\mathbf{x})$ is the image embedding. Correspondingly, $\mathbf{w}_i$ is the text embedding of class y_i obtained by feeding templated texts, e.g., “a photo of a $\{c_i\}$.” into the text encoder. We denote the templated text of class y_i as $\mathbf{t}_i$. Eq. 1 aims to find the most similar text $\mathbf{t}_i$ that maximizes the cosine similarity to the query image. We fully fine-tune the CLIP model for higher performance on downstream tasks, with cross-entropy loss applied to the ground truth and the output distribution.

3.2. Knowledge Distillation on Synthetic Data

CLIP’s zero-shot generalization ability stems from its well learned feature space that aligns images and texts effectively. However, directly fine-tuning on downstream task data distorts this alignment, causing the feature space to collapse into the subspace of specific downstream task and leading to catastrophic forgetting. We resurrect pre-training and historical task data using Stable Diffusion and use these synthetic data as anchors to preserve the integrity of the feature space through distillation, thus mitigating forgetting.

Image Generation. While the optimal distillation data sources to preserve zero-shot capability would be the original pre-training dataset, its unavailability necessitates the use of a diverse generated dataset as an effective approximation. We generate images directly from class names, maintaining a class buffer pool $\mathcal{P}$ that stores all the class names encountered by CLIP in historical downstream tasks, and a base class name set C^0 . Before continual learning begins, $\mathcal{P}$ is initialized with C^0 , a semantically rich class name set used to approximate CLIP’s pre-training data. In particular, we choose the class names of ImageNet [9] as C^0 . As new downstream tasks are learned, their class names are added to $\mathcal{P}$. At the start of task t , $t \geq 1$, we randomly sample class names from $\mathcal{P} = \bigcup_{i=0}^t C^i$ with replacement and template them as “a photo of a $\{c\}$.” to prompt the pre-trained Stable Diffusion to generate images for distillation.

The similarity between CLIP’s and Stable Diffusion’s pre-training data allows images generated by Stable Diffu-

sion to closely approximate CLIP’s pre-training data distribution **without fine-tuning**. Using Stable Diffusion to generate downstream task data also reduces the domain gap between synthetic downstream task data and CLIP’s original pre-training data. Additionally, we ensure inter-class diversity in generated images through the variety of stored class names, resulting in generated images that are widely distributed across CLIP’s feature space. The low generation cost of Stable Diffusion eliminates the need to store generated images. After completing a task, the images are discarded and regenerated before the next task, further ensuring diversity in distillation data sources.

Contrastive Distillation. We implement the distillation loss in a contrastive [19, 39] manner to align with the image-text matching objective of CLIP pre-training. For a synthetic batch containing B image-text pairs at task t , the current model CLIP θ^t encodes this batch into l_2 normalized embeddings $\{(\mathbf{z}_1^t, \mathbf{w}_1^t), (\mathbf{z}_2^t, \mathbf{w}_2^t), \dots, (\mathbf{z}_B^t, \mathbf{w}_B^t)\}$. The image-text similarity within the batch is then computed to obtain the contrastive matrix $M^t = [s_{i,j}^t]_{B \times B}$, where $s_{i,j}^t$ represents cosine similarity $\cos(\mathbf{z}_i^t, \mathbf{w}_j^t)$. Similarly, we use the previous CLIP θ^{t-1} from the last task as the teacher model to compute M^{t-1} . The knowledge distillation loss for image classification is then computed using Kullback-Leibler Divergence [8] to align M^{t-1} and M^t in rows:

$$\mathcal{L}_{KD.image} = -\frac{1}{B} \sum_{i=1}^B M_{i,:}^{t-1} \cdot \log \left(\frac{M_{i,:}^t}{M_{i,:}^{t-1}} \right), \quad (2)$$

where $M_{i,:}^{t-1}$ and $M_{i,:}^t$ denote the i^{th} row of M^{t-1} and M^t respectively. To enhance modality alignment, we compute the text retrieval distillation loss in columns symmetrically:

$$\mathcal{L}_{KD.text} = -\frac{1}{B} \sum_{j=1}^B M_{:,j}^{t-1} \cdot \log \left(\frac{M_{:,j}^t}{M_{:,j}^{t-1}} \right). \quad (3)$$

Symmetrically for both visual and textual modalities, the overall Contrastive Distillation loss is calculated as:

$$\mathcal{L}_{CD} = \mathcal{L}_{KD.image} + \mathcal{L}_{KD.text}. \quad (4)$$

Image-Text Alignment. A challenge in knowledge distillation for CL is that the teacher model, having learned one less task than the student model, also suffers from catastrophic forgetting and can make errors on historical tasks. These errors propagate during the distillation process, causing a worsening modality mismatch in student model. To mitigate this, combining distillation soft targets with the ground truth hard targets is a simple yet powerful solution. Due to Stable Diffusion pre-training [46], the generated images exhibit strong alignment with their corresponding textual prompts in CLIP’s feature space. We use this alignment as hard targets to complement the distillation soft targets,

correcting image-text mismatches caused by errors of the teacher model and thereby ensuring more reliable knowledge retention. To implement this image-text alignment constraint, we calculate KL divergence between the B -class identity matrix I_B (as the alignment matrix) and the contrastive matrix M^t in a manner similar to Eq.2 and Eq.3 respectively, resulting in $\mathcal{L}_{Align.image}$ and $\mathcal{L}_{Align.text}$. Then sum up to get the **Image-Text Alignment loss** $\mathcal{L}_{ITA}$:

$$\mathcal{L}_{ITA} = \mathcal{L}_{Align.image} + \mathcal{L}_{Align.text}. \quad (5)$$

Together with the Cross Entropy loss $\mathcal{L}_{CE}$ for learning new task t , the total training loss with synthetic data-based distillation $\mathcal{L}_{Total}$ can be written as:

$$\mathcal{L}_{Total} = \mathcal{L}_{CE} + \alpha\mathcal{L}_{CD} + \beta\mathcal{L}_{ITA}, \quad (6)$$

where α and β are hyperparameters that balance the trade-off between the terms.

3.3. Adaptive Weight Consolidation

VLMs like CLIP tend to overfit to in-distribution data of specific downstream tasks during fine-tuning, which impairs their ability to generalize to out-of-distribution tasks, including previously learned tasks, thereby exacerbating forgetting. This is even worse when a limited amount of synthetic data fails to maintain the integrity of the feature space with the weakened distillation effect. We introduce adaptive weight consolidation as regularization to alleviate overfitting problems and thus reduce forgetting. To achieve better stability-plasticity balance, we use Fisher information from synthetic image-text pairs during training to adaptively adjust the level of constraints on different parameters.

Weight Consolidation. In practice, the cross-entropy loss of new task learning tends to dominate the optimization process in the early training stages. Specifically, cross-entropy gradients draw the student model parameters towards a sharp local optimum diverging from the teacher model. This divergence leads to in-distribution overfitting and increased distillation loss, making it difficult to return to the broad minimum where the teacher model resides, even after the distillation loss drops and converges. Weight consolidation [66] addresses this issue by introducing an l_2 penalty, which constrains the model parameters to remain close to robust parameters of the teacher model. EWC [31], a typical weight consolidation method, imposes a parameter importance weighted l_2 loss as follows:

$$\mathcal{L}_{EWC} = \sum_i \mathcal{F}_{\theta_i^{t-1}} \cdot (\theta_i^t - \theta_i^{t-1})^2, \quad (7)$$

where the parameter importance $\mathcal{F}_{\theta_i^{t-1}}$ is calculated as the diagonal elements of the Fisher Information Matrix [42].

Adapt to Training Dynamics. Although EWC seeks to maintain model plasticity by applying varying levels of constraints on different parameters based on their estimated

importance, it falls short due to the static nature of this estimation. EWC calculates Fisher information using the old model on the old task. However, since model parameters change dynamically during new task optimization, the Fisher information quickly becomes outdated, resulting in only blind constraints. To address this limitation, we propose adaptive weight consolidation, which keeps updating the Fisher information throughout the optimization process. As optimization progresses, the knowledge distillation loss serves as a reliable indicator of the model’s level of forgetting. Thus, we directly use the distillation loss on the synthetic image-text pairs as the log-likelihood to calculate the diagonal Fisher information:

$$\mathcal{F}_{\theta_i^t}^{(j)} = \left(\frac{\partial (\alpha\mathcal{L}_{KD}^{(j)} + \beta\mathcal{L}_{Align}^{(j)})}{\partial \theta_i^t} \right)^2, \quad (8)$$

where $\mathcal{F}_{\theta_i^t}^{(j)}$ denotes the diagonal Fisher information of model parameter θ_i^t at the j^{th} optimization step. Our adaptive weight consolidation loss is then formulated as:

$$\mathcal{L}_{AWC}^{(j)} = \sum_i \mathcal{F}_{\theta_i^t}^{(j)} \cdot (\theta_i^{t(j)} - \theta_i^{t-1})^2. \quad (9)$$

Notably, $\mathcal{F}_{\theta_i^t}^{(j)}$ is also the squared gradients of the distillation loss, reflecting its stability. When learning new tasks that are significantly different from those previously learned by CLIP, most of the gradient directions of $\mathcal{L}_{CE}$ oppose those of distillation loss. In such cases, $\mathcal{L}_{AWC}$ can constrain parameter updates that are likely to cause drastic changes in distillation loss, i.e., parameter updates that aggravate overfitting and forgetting. This helps smooth out conflicts among multiple optimization objectives and stabilize the distillation loss without compromising plasticity. Our adaptive update of Fisher information utilizes the intermediate results from the back propagation of the distillation loss and introduces minimal computational overhead.

4. Experiments

4.1. Experimental Setting

Datasets. We evaluate our method on two settings: Multi-domain TIL (MTIL) and CIL. MTIL is specifically designed for CL of VLMs and presents a significant challenge as it encompasses 11 datasets with a total of 1,201 classes across various domains, including Aircraft [35], Caltech101 [15], CIFAR100 [33], DTD [7], EuroSAT [22], Flowers [40], Food [5], MNIST [10], OxfordPet [41], StanfordCars [32], and SUN397 [60]. We follow the two-order training protocol proposed in [67] for MTIL. The ablation experiments are conducted on MTIL order I by default. Experiments on the CIL setup are presented in the supplementary material.

Metrics. To evaluate our method on the MTIL setting, we utilize metrics proposed in [67], namely “Transfer”, “Avg.”,

Table 1. Comparison of SOTA methods on MTIL Order I.

Method	Transfer	Δ	Avg.	Δ	Last	Δ
Zero-shot	69.4	-	65.3	-	65.3	-
Continual Finetune	44.6	-	55.9	-	77.3	-
l_2 baseline	61.0	0.0	62.7	0.0	75.9	0.0
LwF [34]	56.9	-4.1	64.7	+2.0	74.6	-1.3
iCaRL [45]	50.4	-10.6	65.7	+3.0	80.1	+4.2
LwF-VR [11]	57.2	-3.8	65.1	+2.4	76.6	+0.7
WiSE-FT [58]	52.3	-8.7	60.7	-2.0	77.7	+1.8
ZSCL [67]	68.1	+7.1	75.4	+12.7	83.6	+7.7
MoE-Adapter [65]	68.9	+7.9	76.7	+14.0	85.0	+9.1
GIFT (Ours)	69.3	+8.3	77.3	+14.6	86.0	+10.1

Table 2. Comparison of SOTA methods on MTIL Order II.

Method	Transfer	Δ	Avg.	Δ	Last	Δ
Zero-shot	65.4	-	65.3	-	65.3	-
Continual Finetune	46.6	-	56.2	-	67.4	-
l_2 baseline	60.6	0.0	68.8	0.0	77.2	0.0
LwF [34]	53.2	-7.4	62.2	-6.6	71.9	-5.3
iCaRL [45]	50.9	-9.7	56.9	-11.9	71.6	-5.6
LwF-VR [11]	53.1	-7.5	60.6	-8.2	68.3	-3.9
WiSE-FT [58]	51.0	-9.6	61.5	-7.3	72.2	-5.0
ZSCL [67]	64.2	+3.6	74.5	+5.7	83.4	+6.2
MoE-Adapter [65]	64.3	+3.7	74.7	+5.9	84.1	+6.9
GIFT (Ours)	65.9	+5.3	75.7	+6.9	85.3	+8.1

and “Last”. The “Transfer” metric assesses the model’s zero-shot capability on unseen data within the task sequence and further reflects the forgetting of pre-training knowledge. “Last” evaluates the model’s ability to retain historical downstream knowledge. “Avg.” is a composite metric assesses the mean performance across “Transfer” and “Last”, reflecting the stability-plasticity balance.

Implementation Details. As in [67], we use the CLIP model with ViT-B/16 [12] as our backbone for all experiments. We train 1K iterations with batch size 64 for each task in MTIL. The introduced hyperparameters α and β are set to 1 and 0.25 respectively to ensure stable results. For image synthesis, we use Stable Diffusion v1.5 [46] with classifier-free guidance set to 7.5 and 50 denoising steps, generating 1K synthetic images per task. More implementation details can be found in the supplementary material.

4.2. Comparison with State-of-the-art Methods

Tab. 1 and Tab. 2 display the performance of different methods on the MTIL benchmark, in order I and order II respectively (detailed in the supplementary material). The different arrangements in order I and order II introduce varying degrees of domain shift. In the Method column, “Zero-shot” denotes the zero-shot performance of the initial CLIP model, indicating the upper bound of the Transfer metric. “Continual Finetune” refers to continual fine-tuning without any protection, suggesting the lower bound of Avg. and

Table 3. Ablation study of different components.

Method	Transfer	Δ	Avg.	Δ	Last	Δ
Zero-shot	69.4	-	65.3	-	65.3	-
Continual Finetune	44.6	-	55.9	-	77.3	-
l_2 Baseline	61.0	0.0	62.7	0.0	75.9	0.0
+CD +ITA +AWC	Transfer	Δ	Avg.	Δ	Last	Δ
✓	✓		63.5	+2.5	70.5	+7.8
✓	✓		68.3	+7.3	76.3	+13.6
✓	✓		68.9	+7.9	76.6	+13.9
✓		✓	68.7	+7.7	77.0	+14.3
✓	✓	✓	69.3	+8.3	77.3	+14.6
					86.0	+10.1

Last metrics. We also introduce a straightforward baseline, referred to as the “ l_2 baseline”, which employs a direct, un-weighted l_2 penalty to prevent significant changes in model parameters before and after learning a new task. Previous methods [11, 34, 45, 58] have shown limited improvement in the Last metric, and their Transfer metric performance is significantly lower compared to the l_2 baseline, as most of them don’t take into account the forgetting of pre-training knowledge. We attribute the non-trivial performance of the l_2 baseline, particularly in MTIL order II with its significant domain shift, to the fact that the optimum reached by CLIP through pre-training is broad and flat. Our proposed GIFT outperforms all other methods across all metrics, including the strong competitors ZSCL [67] and MoE-Adapter [65]. MoE-Adapter introduces extra parameters that grow linearly but performs worse when the domain shift is drastic (i.e., in MTIL Order II). It’s worth nothing that our method using 1K synthetic images outperforms ZSCL using 100K ImageNet images, demonstrating the effectiveness of synthetic data in balancing between preserving general knowledge and retaining downstream task learning.

4.3. Ablation Study of Different Components

In this section, we perform ablation studies on our proposed method. We decompose our approach into three components: Contrastive Distillation (CD), Image-Text Alignment (ITA), and Adaptive Weight Consolidation (AWC). CD and ITA together constitute the synthetic data-based distillation. Each component is incrementally added to the l_2 baseline, and the experimental results are summarized in Tab. 3. The performance of using only ITA is significantly worse than using only CD when both are scaled to 1. This underscores the advantage of distillation’s soft targets over complete hard targets in terms of memory retention. Soft targets encapsulate more latent information that can reflect learned knowledge, and thus is more robust to the possible noise in synthetic data. When the scales of CD and ITA are set to 1 and 0.25 respectively, the Transfer performance improves by 0.6%. This suggests that, in the right proportion, hard targets can correct erroneous information in soft targets provided by the teacher model, ensuring accurate knowledge retention. Finally, as an enhanced version of l_2 weight

Table 4. **Analysis of distillation mechanism.** The default settings are marked in gray, which employs a contrastive distillation loss, the last CLIP model as the teacher model, and $\beta = 0.25$ for ITA scale.

(a) Distillation Loss.				(b) Teacher Model.				(c) Scale of Image-Text Alignment.			
Loss	Transfer	Avg.	Last	Teacher	Transfer	Avg.	Last	ITA Scale	Transfer	Avg.	Last
Feat. Dist.	64.0	71.6	80.5	Initial CLIP	69.1	74.0	80.1	$\beta = 0.0$	68.3	76.3	84.7
Image-only	66.8	75.1	84.1	Last CLIP	68.9	76.6	85.0	$\beta = 0.25$	68.9	76.6	85.0
Text-only	64.7	71.9	81.8	WiSE(0.2)	69.1	76.1	83.4	$\beta = 0.5$	68.7	76.2	84.2
Contrastive	68.9	76.6	85.0	WiSE(0.5)	69.6	75.3	81.6	$\beta = 1.0$	68.5	75.4	82.4

consolidation, AWC is employed to dynamically adjust the constraint level during training. This adaptive approach enhances both the stability and plasticity of the model, leading to overall improvements across all three metrics.

4.4. Analysis of Synthetic Data-based Distillation

In this section, we conduct ablation studies and analysis on the synthetic data-based distillation. To isolate the effect of AWC, we apply CD and ITA to the l_2 baseline. The analysis primarily focuses on three aspects of distillation (objective function, teacher model and data source), along with the scale of our proposed complementary hard target ITA.

Distillation Loss. Tab. 4 (a) compares several types of distillation loss. The feature distance loss [71], penalizes l_2 distance of the visual and textual embeddings between the teacher and student models separately, performs poorly due to lack of consideration for modality alignment. We also implement image-only and text-only distillation by excluding one of the two encoders from the loss computation in the student model, as introduced in [67]. Since both encoders suffer from forgetting, these two underperform compared to our contrastive distillation loss.

Teacher Model. Tab. 4 (b) shows the distillation performance with different teacher models. Utilizing an initial CLIP that hasn’t been trained on downstream tasks yields higher Transfer values. However, emulating the output of such a model weakens the memory of downstream tasks. Using WiSE-FT [58] to linearly interpolate between the initial CLIP and the previous CLIP also enhances Transfer performance but still fails to achieve a good balance with learning new tasks and retaining historical tasks.

Scale of Image-Text Alignment. Tab. 4 (c) illustrates the distillation performance across varying scales of ITA, with the α value fixed to 1. A smaller β balances the soft and hard targets well and corrects errors in the soft targets. While a larger β can cause the model to overfit the synthetic samples, impeding model’s ability to learn new tasks.

Number of Synthetic Images. Fig. 3 presents the ablation study on the number of synthetic images generated for distillation per task. To reduce instability caused by varying generated images, we use a consistent seed setting, ensuring smaller image sets are subsets of larger ones. The results show that increasing the number of synthetic images broadens the distribution of anchor points in CLIP’s fea-

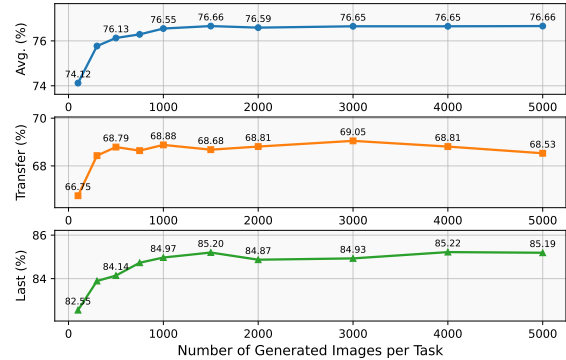


Figure 3. Generating different numbers of synthetic images as distillation data sources for each task produces different results.

Table 5. **Comparison of data sources for distillation.** “Synthetic” represents the default setting, which uses both synthetic pre-training and synthetic downstream task data. “Synthetic†” uses only synthetic pre-training data, i.e., a synthetic ImageNet.

Method	Source	Num	Transfer	Avg.	Last
GIFT	ImageNet	1K	69.0	76.0	84.0
	Synthetic		68.9 (-0.1)	76.6 (+0.6)	85.0 (+1.0)
	Synthetic†		68.3 (-0.7)	75.5 (-0.5)	83.7 (-0.3)
GIFT	ImageNet	3K	69.3	76.6	84.7
	Synthetic		69.1 (-0.2)	76.7 (+0.1)	84.9 (+0.2)
	Synthetic†		69.0 (-0.3)	75.9 (-0.7)	84.1 (-0.6)
ZSCL	ImageNet	100K	68.1	75.4	83.6

ture space, thereby improving distillation effect. However, performance gains plateau beyond a certain scale (at 1K images). Therefore, we conclude that generating 1000 images is both an economical and effective choice.

Comparison with Real Images. Tab. 5 presents the experimental results of replacing synthetic images with an equivalent number of real images sampled from ImageNet [9] for each task. With a limited allowance of 1K images, using synthetic images yields higher Avg. and Last values compared to ImageNet images, while maintaining a comparable Transfer value. Because ImageNet images are closer to CLIP’s pre-training data and differ in domain from certain downstream tasks, replaying ImageNet data exacerbates forgetting of these tasks, resulting in a lower Last value despite the higher Transfer score. Using synthetic images offers the advantage of balancing the retention of both pre-training and downstream task knowledge. To illustrate this, we generate data solely based on ImageNet la-

Table 6. **Comparison of AWC and EWC variant.** “EWC-V” refers to the variant of EWC which uses static parameter importance estimation based on synthetic data.

Method	Num of Samples	Transfer	Avg.	Last
EWC-V	64	65.8	75.6	85.6
EWC-V	128	66.8	76.1	85.9
EWC-V	256	68.0	76.8	86.2
AWC	-	69.3	77.3	86.0

bels (“Synthetic+”), resulting in performance significantly lower. This outcome also reflects the issue of greater noise in synthetic images compared to real ImageNet data.

4.5. Analysis of Adaptive Weight Consolidation

We ablate and analyze our adaptive weight consolidation method on synthetic data-based distillation without l_2 constraints. The results are shown in Fig. 4, Fig. 5 and Tab. 6.

Effect on Distillation Loss. To better represent the distillation loss, we visualize the cross-entropy between the output distributions of the teacher and student models instead of the KL divergence in Fig. 4, i.e., $H(M^{t-1}, M^t) = D_{KL}(M^{t-1} \parallel M^t) + H(M^{t-1})$. The reason is that KL divergence primarily measures the alignment with the teacher model’s distribution, which may not accurately reflect the effectiveness of distillation in mitigating forgetting when the teacher model is suboptimal. In Figure 4, a higher initial distillation loss at every 1Kth iteration indicate a poorer teacher model. By incorporating AWC, we can achieve a better teacher model and improved distillation performance.

Effect on Combating Overfitting. Fig. 5 illustrates the impact of AWC on combating overfitting by visualizing the train loss on the Aircraft dataset and the test loss on unlearned CIFAR100. Fine-tuning the initial CLIP model on Aircraft, without and with AWC, yields two distinct local optima W_1 and W_2 , where W_2 is flatter and broader than W_1 . Meanwhile, for the test loss on CIFAR100, which reflects the model’s zero-shot capability, W_1 falls into a high-loss region, whereas W_2 remains within the broad minimum W_0 of the initial CLIP model. AWC guides the optimization process to find the overlapping area between the current and historical optimization target within broad minimum of initial CLIP, avoiding falling into the sharp local minimum trap and thus alleviating overfitting.

Comparison with EWC. To verify the effectiveness of AWC in dynamically estimating and updating parameter importance, we implement a variant method similar to EWC [31] that uses static parameter importance estimation. In its original design, EWC offers no protection during the first task, which poses a significant disadvantage for VLMs with pre-training knowledge. We develop an EWC variant that extracts several batches of synthetic data before each task begins to compute Fisher information as a static parameter importance weight. This weight remains fixed throughout

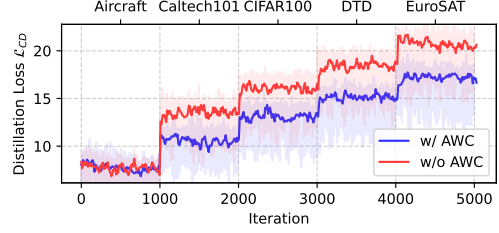


Figure 4. **CD loss for the first 5 tasks in MTIL order I.** In our implementation, cross-entropy is used as equivalent instead of KL divergence to compute $\mathcal{L}_{CD}$, and the results are presented accordingly. Lower loss means better mitigation of forgetting.

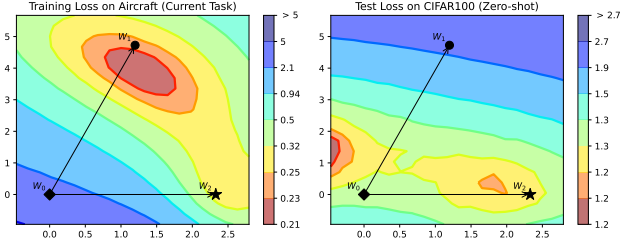


Figure 5. **Loss values on a two dimensional slice of the loss landscapes.** We use W_0 , W_1 and W_2 to represent the initial CLIP and CLIP models finetuned on the Aircraft [35] dataset without and with AWC, respectively. As in [18], we obtain an orthonormal basis u_1 , u_2 for the plane spanned by these models, and the x and y-axis show movement in parameter space in these two directions.

the training process of a downstream task. As shown in Tab. 6, increasing the number of samples used to compute the static parameter importance improves the performance of this EWC variant. However, it still falls short of AWC, which better accounts for the training dynamics.

5. Conclusion

In this paper, we propose GIFT, a novel continual fine-tuning approach for VLMs that utilizes diffusion-generated data to mitigate forgetting. GIFT includes a distillation process that encourages the VLM to revisit learned knowledge on matching synthetic image-text pairs and an adaptive weight consolidation strategy for regularization. Since the introduction of synthetic data recreates both pre-training and downstream data without storage and privacy issues, we achieve a balance between preserving general knowledge and retaining downstream task learning in VLMs. Furthermore, our results show that distillation with l_2 constraints yields improved performance, allowing stable results with less synthetic data generated. Fisher information from synthetic data enables adaptive l_2 constraint, achieving a better stability-plasticity balance. Comprehensive experiments exhibit superior performance compared to the SOTA.

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